

Dennis Krasnov

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Skills

Backend: Java, Go, NodeJS, GraphQL, gRPC, PostgreSQL, SQLite, Redis, Kafka

Frontend: HTML, CSS, JavaScript, TypeScript, React, HTMX

System: Rust, Zig, C, Linux, eBPF, io_uring, networking, benchmarking

Infrastructure: Bash, Docker, Kubernetes, HAProxy, Terraform

Architecture: DDD, gathering requirements, diagrams, documentation

Work

Backend Software Engineer, *Power TakeOff* Dec 2023 - Present

- Created flexible GraphQL-like endpoint for querying a complicated hierarchy of business entities. Reduced technical debt by eliminating 4 legacy endpoints using the strangler fig pattern. Introduced Testcontainers and 178 regression tests.
- Re-implemented TimescaleDB query that filters, upsamples, and aggregates terabytes of time series data in PL/pgSQL. Enables efficient on-demand derivation of a cost (\$) time series from a consumption (kWh) and rate (\$/kWh) time series.
- Significantly optimized algorithm that detects electricity meters renamed or moved in the hierarchy. Avoided expensive cartesian product by aggressively filtering out incompatible candidates based on their position.
- Designed and implemented API keys with fine-grained scopes that integrate with the existing JWT-based auth.
- Led conditional alerts project that sends email alerts based on the result of daily analyses. Designed a horizontally scalable service that performs recurring analyses of electricity consumption data. Team of 4 worked on project for 1.5 months.
- Delivered 3 hour-long presentations at internal developer conferences. Topics: concurrency, HTTP, API security.

Backend Software Engineer Intern, *Dapper Labs* May 2022 - Aug 2022

- Optimized a complex product listings SQL query. Implemented the join and kick endpoints for the squads service.
- Removed unnecessary CQRS pattern from microservice template, saving 1000+ lines of boilerplate in the squads service.

Full-Stack Software Engineer Intern, *WFHome* May 2021 - Aug 2021

- Automated shipping label generation using the Canada Post API. Slashed typical order-processing labor from 55h to 2h.
- Split up login page into sign in and sign up. Created multi-step onboarding which resolved 12 backlog tickets.

Backend Software Engineer Intern, *CGI* Jan 2021 - Apr 2021

- Migrated Spring services from RabbitMQ to Solace PubSub. Deployed Consul on RHEL servers using templated systemd.

Backend Software Engineer Intern, *Ultimate Software* Sep 2019 - Dec 2019

- Added permissions claim to JWT on ForgeRock identity gateway serving 51M users. Solved zero-trust chicken-egg issue.

Freelance Software Engineer, *Krasnov Tech* Jul 2016 - Aug 2019

- Planned, designed, developed, released, and maintained two products. sgbiotec.com, a store selling lab-grade fluorescent proteins using React, Shopify API. Pocket Trade Show, a marketplace platform for salesmen using PHP, MySQL, AWS S3.

Education

University of Waterloo, *Bachelor of Computer Science* 2018 - 2023

- Courses: data structures, algorithms, operating systems, compilers, networking, databases, distributed systems, security.

Projects

testpropeller.com: Restores Docker containers from snapshots in milliseconds using gVisor. CLI routes requests to nearest scheduler and tunnels TCP/UDP traffic. Multi-tenant dashboard manages credit-based billing with top-ups and subscriptions.

Gibe: Zig HTTP 1.1 server library. Zero memory allocations using virtual-memory-backed ring buffers and arena allocators.

Uringy (27★): Rust structured concurrency library. Issues non-blocking syscalls in batches using io_uring. Userspace virtual threads spawn 15x and context switch 22x faster than Tokio tasks. Supports leak-free cascading cancellation.

Deep Link Navigation (68★): Flutter navigation library. Elegant URL-like abstraction, well documented and tested.

Neural Network Driving: Autonomous driving AI trained with reinforcement learning. Realtime and interactive visualization. Won gold at Canada-Wide Science Fair, flown to Ottawa to present to prime minister Justin Trudeau.